

GURMANDER SINGH MAAN

AI/ML Engineer / Software Developer

Vancouver, BC, Canada · Permanent Resident

Gurmandersinghmaan@gmail.com · github.com/Gurmander

SUMMARY

AI/ML Engineer and Software Developer completing a BS in Data Science and Applications at IIT Madras, building ML-enabled applications and backend systems with Python, PyTorch, FastAPI, and PostgreSQL. Experience spans deep learning, computer vision, REST APIs, geospatial optimization, and CI/CD, with a **Best Runner-up** finish at the IIT Madras × Syngenta Hackathon.

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, SQL
AI / ML	PyTorch, Scikit-learn, Hugging Face, CNNs, Transformers, NumPy, Pandas, Librosa, Torchaudio
Backend / Data	FastAPI, Flask, REST APIs, SQLAlchemy, Pydantic, PostgreSQL, Redis, Celery
Frontend / DevOps	React, Vue.js, Docker, GitHub Actions, Git, Linux, Cloudflare R2

PROJECTS

Syngenta Field Force Intelligence

May 2026

IIT Madras × Syngenta Hackathon · **Best Runner-up** · FastAPI · React · OR-Tools

- Implemented Google OR-Tools route optimization with OSRM road-distance matrices and route geometry, including geospatial and coordinate-processing workflows for multi-stop field routes.
- Contributed to AI voice/route-analysis features and GitHub Actions CI/CD, including automated testing and deployment of the backend to Hugging Face and frontend to Netlify.

Messy Mashup - Music Genre Classification

Mar 2026

PyTorch · Librosa · Torchaudio · W&B

- Built a 10-class audio-classification pipeline for noisy, distribution-shifted data using mel-spectrograms, delta features, SpecAugment, synthetic mashups, and ESC-50 environmental-noise augmentation.
- Trained and compared CNN, CNN-Transformer, and Audio Spectrogram Transformer architectures; tracked experiments with W&B and achieved **0.945 Macro F1**, ranking **86th of 1,119 participants (top 8%)**.

Urban Fix - AI-Powered Civic Service Platform

Jun 2026 – Aug 2026

FastAPI · PostgreSQL · Vue.js · YOLOv11 · Gemini Vision · Docker

- Developed backend and AI workflows for role-based civic complaints, including geolocation-based ward assignment, lifecycle tracking, and photo-based verification using FastAPI, SQLAlchemy, PostgreSQL, and JWT authentication.
- Integrated YOLOv11 for pothole/garbage detection and vision models for broader issue analysis; added GitHub Actions CI for automated unit/integration tests and Ruff linting on pushes and pull requests.

EXPERIENCE

TELUS

Jan 2023 – Jan 2024

SmartHome Security Customer Support

Surrey, BC

- Resolved device, connectivity, and setup issues for residential SmartHome Security systems, translating technical diagnostics into clear customer guidance in a customer-facing support environment.

TELUS

2020 – Oct 2022

Install & Repair Technician

Surrey, BC

- Installed and troubleshooted residential internet/Wi-Fi, television, and phone services onsite, diagnosing network and hardware issues and completing service work independently.

EDUCATION

Indian Institute of Technology Madras

Apr 2024 – Apr 2027 (Expected)

BS in Data Science and Applications

GPA: 9.5/10

Relevant Coursework: Deep Learning, Machine Learning, Software Engineering, Software Testing, Database Management Systems, AI: Search Methods.

Langara College - Vancouver, BC

Jan 2017 – Jul 2019

Associate of Science

Dean's Honour Roll